Exercises 7

7.1 Substitution

Evaluate the integral.

1. $\displaystyle\int 2x\cos(x^2)\,dx$

2. $\displaystyle\int_0^1 x\sqrt{1-x^2}\,dx$

3. $\displaystyle\int \frac{e^x}{1+e^x}\,dx$

4. $\displaystyle\int_0^{\pi/2} \sin^3 x\,\cos x\,dx$

7.2 Integration by Parts

Evaluate the integral.

5. $\displaystyle\int xe^{2x}\,dx$

6. $\displaystyle\int_1^e x^2\ln x\,dx$

7. $\displaystyle\int e^x\sin x\,dx$

8. (a) Show that for every positive integer n, $\displaystyle\int x^n e^x\,dx = x^n e^x - n\int x^{n-1}e^x\,dx$. (b) Use part (a) to evaluate $\int_0^1 x^3 e^x\,dx$.

7.3 Partial Fractions

Evaluate the integral.

9. $\displaystyle\int \frac{dx}{x^2-1}$

10. $\displaystyle\int \frac{3x+5}{(x+1)(x+2)}\,dx$

11. $\displaystyle\int_0^1 \frac{dx}{(x+1)(x+2)}$

7.4 Improper Integrals

12. For which values of p does the integral $\displaystyle\int_1^\infty \frac{dx}{x^p}$ converge? Find its value when it does.

13. Evaluate $\displaystyle\int_0^\infty xe^{-x}\,dx$.

14. The region under the curve $y = 1/x$ for $x \ge 1$ is rotated about the x-axis. Show that the resulting solid has finite volume although the region itself has infinite area.

Exercises and Problems

Section 4.1 Forces in One and Two Dimensions

1. Two forces act on a hockey puck resting on frictionless ice, as shown in **FIGURE P4.1**. Force F_1 has magnitude 3.0 N and points east; force F_2 has magnitude 4.0 N and points north. Find the magnitude and direction of the net force on the puck.

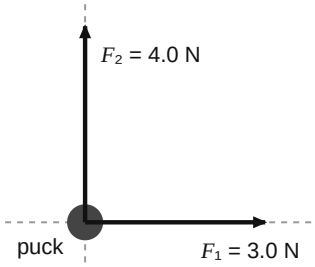


FIGURE P4.1

2. A 1200 kg car travelling at 25 m/s brakes to a stop in 5.0 s with constant acceleration. What is the magnitude of the net force on the car while it is braking?

3. A 200 N sign hangs at rest from two cables, as shown in **FIGURE P4.3**. The left cable makes an angle of 30° with the ceiling and the right cable makes an angle of 45°. Find the tension in each cable.

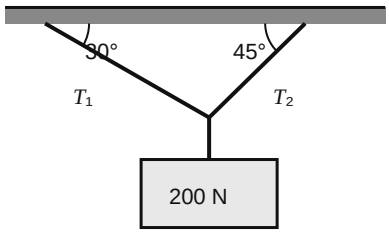


FIGURE P4.3

Section 4.2 Newton's Second Law

4. A 2.0 kg block is released from rest on a frictionless incline that makes an angle of 30° with the horizontal, as shown in **FIGURE P4.4**. (a) What is the acceleration of the block? (b) How fast is it moving after sliding 1.5 m along the incline?

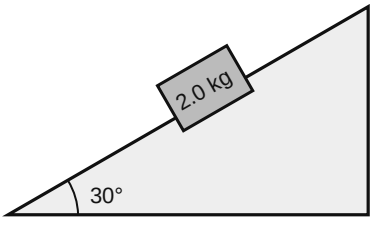


FIGURE P4.4

5. An elevator of mass 850 kg accelerates upward at 1.2 m/s². What is the tension in the cable that supports it?

6. Two blocks of mass 3.0 kg and 5.0 kg are connected by a light string that passes over a frictionless, massless pulley, as shown in **FIGURE P4.6**. The blocks are released from rest. Find (a) the acceleration of the blocks and (b) the tension in the string.

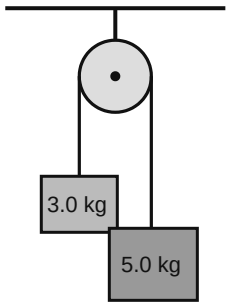


FIGURE P4.6

Section 4.3 Motion Graphs

7. **FIGURE P4.7** shows the velocity-versus-time graph of a cyclist moving along a straight road. (a) What is the cyclist's acceleration between t = 0 s and t = 2 s? (b) How far does the cyclist travel between t = 0 s and t = 6 s?

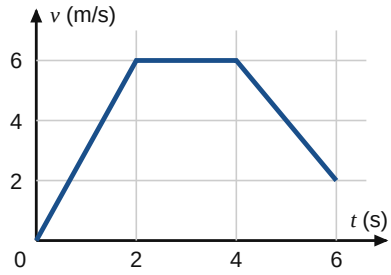


FIGURE P4.7

8. A ball is thrown horizontally at 12 m/s from the top of a 20 m tall building. Ignoring air resistance, how far from the base of the building does it land?

Section 4.4 Friction

9. A 15 kg crate rests on a horizontal floor. The coefficient of static friction between the crate and the floor is 0.40. What is the smallest horizontal force that will start the crate moving?

10. A 10 kg sled is pulled across level snow by a rope that makes an angle of 25° with the horizontal, as shown in **FIGURE P4.10**. The tension in the rope is 40 N and the coefficient of kinetic friction is 0.10. Find the acceleration of the sled.

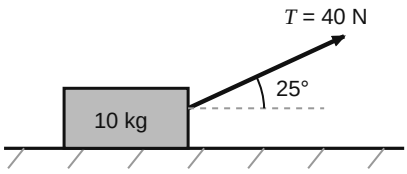


FIGURE P4.10

11. A car travelling at 20 m/s on a level road locks its brakes and skids to a stop. If the coefficient of kinetic friction between the tyres and the road is 0.70, how far does the car skid?

Section 4.5 Circular Motion

12. A 0.20 kg ball on the end of a 0.80 m string is whirled in a horizontal circle at a constant speed of 4.0 m/s. What is the tension in the string? Ignore gravity.

13. A small bob of mass *m* hangs from a string of length 1.2 m and moves in a horizontal circle, so that the string makes a constant angle of 20° with the vertical, as shown in **FIGURE P4.13**. Find the speed of the bob.

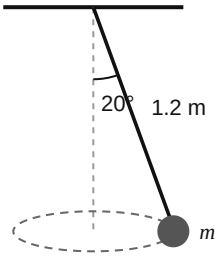


FIGURE P4.13

14. A 1000 kg car rounds a flat curve of radius 50 m. If the coefficient of static friction between the tyres and the road is 0.80, what is the greatest speed at which the car can take the curve without sliding?